

9. Popcorn lung is a lung disease which results in scarring and inflammation of the bronchioles. Symptoms of the disease are coughing and difficulty breathing. **Image 9.1a and 9.1b** show a healthy bronchiole and a bronchiole affected by popcorn lung.

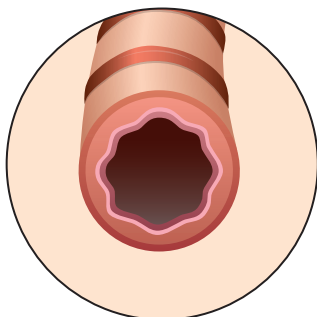


Image 9.1a – healthy bronchiole

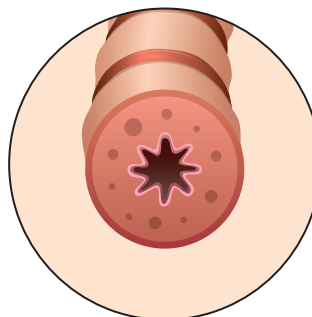


Image 9.1b – bronchiole affected by popcorn lung

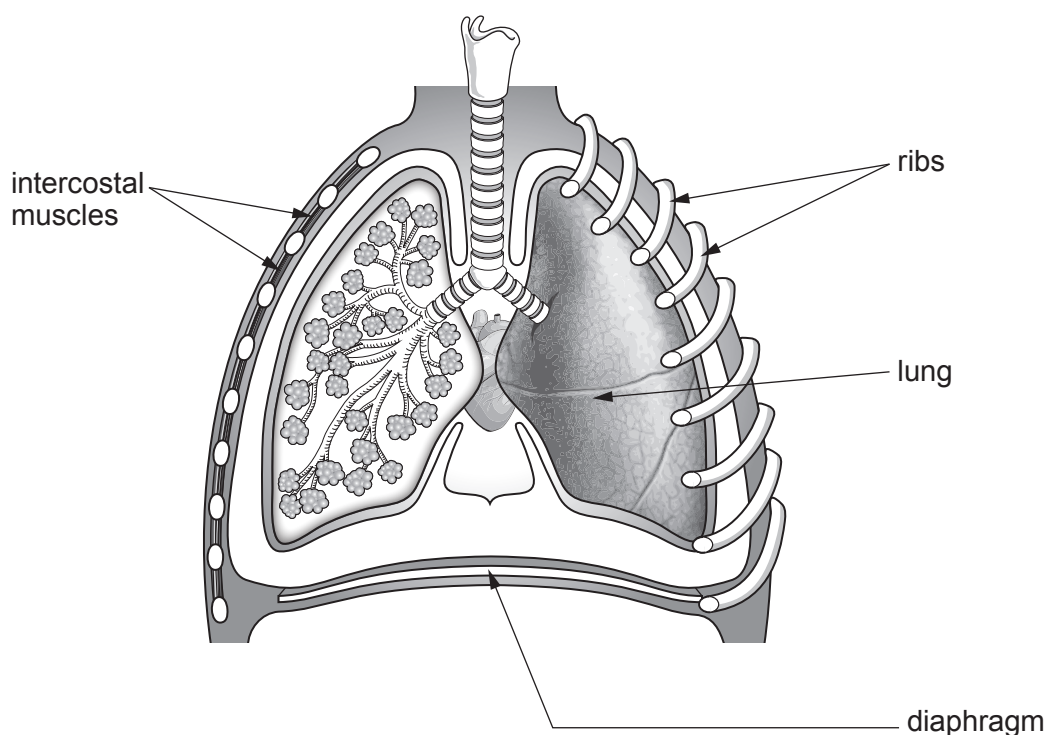
A chemical called diacetyl is used in the manufacture of popcorn. A link between inhaling diacetyl and the lung disease was made when a group of workers from a popcorn factory was found to have developed the disease.

In the US there have been increasing numbers of cases of popcorn lung linked to the use of e-cigarettes. There are around 7 700 flavours for e-cigarettes sold in the US. A single study of 51 e-cigarette flavours in 2016 found that 39 contained diacetyl. Diacetyl was banned in e-cigarette flavourings in the UK in 2016.

In 2019, a report in the journal *Nature* showed that diacetyl reduced the number of ciliated cells in the airways.

- (a) **Image 9.2** shows the human respiratory system.

Image 9.2



Examiner
only

(i) Use a **labelled arrow** to identify an alveolus on **Image 9.2**. [1]

(ii) State **two** ways in which the alveolus is adapted for gas exchange. [2]

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(b) Explain why a person affected by popcorn lung shows the following symptoms:

(i) coughing. [1]

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(ii) breathing difficulties. [2]

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(c) Use the information given to evaluate the validity of the following conclusion: [2]
'E-cigarette flavours cause popcorn lung'.

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END OF PAPER

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